

Evaluating Composable Verification via Magic-Blindness

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Motivation

Cloud quantum computing delegates computation to remote hardware that may be untrusted. Clients therefore need both privacy and assurance that their computation was performed correctly. We benchmark a recent protocol that brings composable, noise-robust, trap-based verification to the circuit model used by most quantum architectures [1].

Background

Clifford only circuits

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Gottesman-Knill[2]

Classically simulable circuits

Clifford + MSI

Universal quantum computation

In *Magic-Blindness*, a computation round with MSIs and a test round with stabilizer inj. are indistinguishable to the server, allowing a client to design classically simulable deterministic trap rounds to detect a cheating server.

Protocol Summary

Compile into Clifford + MSI

Randomly order trap and comp. rounds

Send blinded (QOTP) input to server.

Trap majority vote

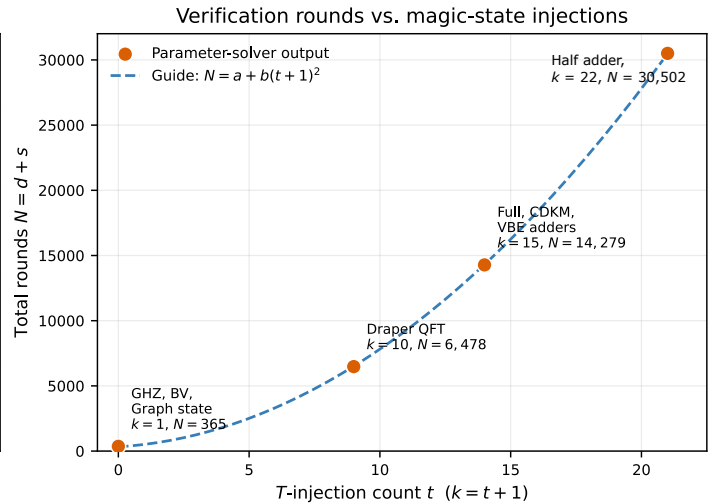
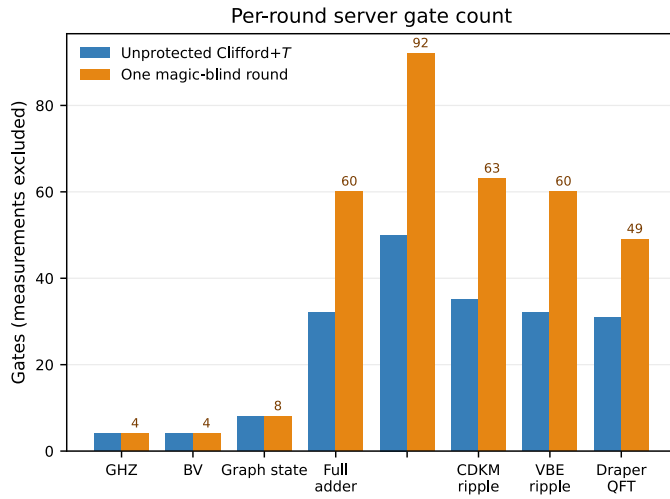
The client begins by compiling the circuit into Clifford + MSI layers. Then, the client will secretly order comp. and trap rounds for the server. On any round, the client blinds their input using quantum one-time pads (QOTP), randomly applying Pauli X,Z gates to all qubits and a Z rotation of a multiple of $\pi/2$ to the injection target qubit. Then, the server applies *CNOT*, *SWAP* for injection and a Z rotation. The angle of the rotation is passed to the server by the client and determined by the QOTP rotation and if there is a measurement correction. Since the server does only these gates at any injection, the server cannot distinguish a trap from a computation round.

On a trap round's injection, a client will inject a precomputed stabilizer state instead of a magic. For instance, if we wanted the deterministic outcome on the i th qubit to be 0, the input state would be the +1 eigenstate of $C^\dagger Z_i C$, where C is the magic-free circuit. Additionally, the following bounds on the number of comp. (d) and test (s) rounds can be derived from the correctness and security guarantees in the paper (Lemma 5 and theorem 6), with security param. ϵ and inj. count .

$$d = O(\log 1/\epsilon)$$

$$s = O((t+1)^2 \log \frac{1}{\epsilon})$$

Magic-Blind Verification Overhead on MQT Benchmarks



Benchmarks

A modular resource estimator compiles each MQT circuit [3] to Clifford+T, counts magic-state injections (t), sets ($k=1+t$), and iteratively solves the paper's bounds for (s,d,w) . GHZ, Bernstein-Vazirani, and graph-state circuits contain only Clifford gates, so ($t=0$), their per-round blinding overhead is negligible, and they require the fewest verification rounds. For non-Clifford benchmarks, per-round MSI overhead remains moderate, but total cost is dominated by verification repetitions; circuits with the same T-count consequently have close round counts. Several additional benchmarks were excluded because arbitrary-angle synthesis produced prohibitively large T-counts. These results are resource estimates using one designated decision wire, rather than full multi-output verification.

Future Work

Trap merging: Traps can share a test round when their propagated stabilizers admit a common tensor-product eigenstate. By constructing a graph with edges in between incompatible traps, trap merging can be reduced to graph coloring (NP-Hard) [4]. Heuristic coloring could nevertheless reduce the effective trap-family count k and therefore the number of test rounds.

CCZ Injection: Circuits like mod. exponentiation, Grover's, etc. and corresponding architectures are Toffoli-based (ex. Gidney's Shor architecture [5]). Extending the guarantees of magic-blindness from T to CCZ injections would reduce Clifford QOTP overhead and may reduce number of traps (and thus rounds) by reducing number of injections. Establishing this requires new CCZ blindness, correction-bookkeeping, and trap-detection proofs; the resulting reduction in k is therefore conjectural.